STORE SALES DATASET ANALYSIS

Project Documentation

Prepared for the DEPI Project

Team Members

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Dataset: Superstore Sales Dataset.csv

This document contains only the three requested sections.

1. Project Planning & Management

1.1 Project Idea

The project analyzes the supplied Superstore sales transaction dataset and uses Microsoft Power BI to present sales performance in an understandable, interactive way. The analysis will focus on sales over time, product category and sub-category performance, regional patterns, customer segments, and shipping methods. The project is intended to support evidence-based business decisions rather than make claims that the available data cannot support.

1.2 Project Objectives

- Understand the structure and quality of the supplied sales dataset.
- Summarize total sales and compare sales across dates, regions, categories, sub-categories, and customer segments.
- Identify high-performing and low-performing sales groups using the available Sales field.
- Build a clear Power BI report with suitable KPIs, charts, and filters.
- Document the project plan, relevant existing solutions, stakeholders, and system requirements.

1.3 Project Scope

In scope:

- Inspecting and preparing the CSV data for analysis, including checking missing values, duplicate records, and data types.
- Creating descriptive sales measures and visuals using fields available in the supplied file.
- Analyzing sales by order date, category, sub-category, region, state/city, customer segment, and ship mode.
- Designing an interactive Power BI report and presenting findings in a final project report.

Out of scope / data limitations:

- Profit, cost, quantity, and discount analysis, because these fields are not present in the supplied CSV.
- Causal claims about why sales changed; the dataset supports descriptive comparisons, not proof of causation.
- Forecasting or predictive modeling unless separately approved and supported by additional data.

1.4 Dataset Snapshot

Item	Observed value
Rows / columns	9,800 rows / 18 columns
Order date coverage	03 Jan 2015 to 30 Dec 2018
Distinct orders	4,922
Distinct customers	793
Distinct products	1,861
Categories / sub-categories	3 / 17
Regions / states / cities	4 / 49 / 529
Total sales (sum of Sales)	2,261,536.78 (dataset currency not specified in file)
Fully duplicated rows	0
Missing Postal Code values	11

The file contains 18 fields, including Order Date, Ship Date, Ship Mode, Customer ID, Segment, Region, Category, Sub-Category, Product Name, and Sales. All rows show the United States as the country in the inspected sample. The 11 missing Postal Code values should be reviewed; they should not be filled with invented values.

1.5 Timeline / Gantt Plan

Work package	Week 1	Week 2	Week 3	Week 4
Dataset understanding and preparation	●			
Define analysis questions and KPIs		●		
Build Power BI report and validate visuals			●	
Final documentation and presentation				●

The schedule follows the four-week project plan provided by the team. Exact calendar dates can be added if the trainer requires dated milestones.

1.6 Task & Responsibility Allocation

Team member	Proposed responsibility
Kareem Ahmed Abdel Moneim Abdel Mageed	Coordinate the work plan, maintain documentation, and support analysis validation.
Ahmed Abdel Rabbo Hassan Abdel Rabbo	Inspect the dataset, lead data cleaning/preprocessing, and document data-quality decisions.
Salah Mohamed Salah Ali	Develop business questions, support calculations, and interpret analytical results.
Mahmoud Yasser Mahmoud Mohamed	Build and test dashboard visuals, filters, layout, and support the final presentation.

This allocation is a proposed starting point and should be confirmed by the team before submission.

1.7 Risks & Mitigation

Risk	Mitigation
Missing values (11 in Postal Code)	Inspect affected records; retain blanks or use a documented, justified treatment rather than inventing values.
Incorrect date parsing	Set and verify the date format in Power Query; validate minimum and maximum dates after conversion.
Unavailable measures	Do not report profit, quantity, or discount insights because these fields are absent from the supplied file.
Misleading comparisons	Use consistent filters and aggregation; label KPIs clearly and validate totals against the source data.
Timeline delays	Assign an owner to each task, review progress weekly, and complete a basic working report before adding optional visuals.
Privacy exposure	Avoid displaying unnecessary customer names or IDs in public screenshots and shared presentations.

1.8 KPIs / Measures of Success

- Total Sales = SUM of the Sales column.
- Number of Orders = distinct count of Order ID.
- Average Sales per Order = Total Sales divided by distinct Order ID count (define and document the calculation consistently).
- Sales by Category, Sub-Category, Region, Segment, and Ship Mode.
- Monthly or yearly sales trend based on the parsed Order Date.
- Report quality: all key visuals respond correctly to slicers, totals reconcile with the source, and the final report is delivered within the four-week plan.

2. Literature Review

The review considers existing Superstore sales dashboards and official Microsoft guidance. These sources are used as design and technical references, not as proof that our project has already achieved the same outcomes.

Atul1804, Superstore Sales Dashboard (GitHub)

Describes a Power BI Superstore project with sales overview, customer analysis, and profit analysis. Useful as a reference for report structure and interactive filtering. Its listed fields include Profit and Quantity, which are not present in our supplied CSV, so those measures cannot be reproduced from our current dataset.

Reference: https://github.com/atul1804/Superstore-Sales-Dashboard

Mzaeemhaider, Superstore Sales Dashboard Power BI (GitHub)

Describes an interactive Power BI project focused on sales trends, customer insights, regional performance, and executive KPIs. Relevant to our available date, customer, region, category, and sales fields.

Reference: https://github.com/mzaeemhaider/superstore-sales-dashboard-powerbi

2.1 Comparison: Strengths and Limitations

Reference / approach	Useful strengths	Limitations for our project
Existing Superstore Power BI dashboards	Demonstrate familiar ways to organize sales overview, customer, category, regional, and time-based views; show how KPIs and filters can be combined.	Dashboard designs vary in detail and quality. Some reference datasets include Profit, Quantity, or Discount, which are not included in our CSV. Those analyses cannot be copied directly.
Official Power BI documentation	Provides product-specific guidance for reports, data modeling, and analysis.	Documentation is general guidance, not a ready-made analysis of this exact dataset.
Official Power Query documentation	Explains data import and transformation practices that can be applied during preparation.	It does not decide how missing Postal Code values should be handled in our particular records; that requires a documented project decision.

2.2 What Is Different / Useful About Our Project

- The analysis is explicitly limited to the 18 columns in the supplied CSV and avoids unsupported profit, quantity, or discount claims.
- The report can combine sales trends with region, product category, customer segment, and shipping-method views using fields available in the data.

- The project documents a transparent dataset-quality check, including the 11 missing Postal Code values and zero fully duplicated rows found in the initial inspection.
- The dashboard is intended to be practical for a reviewer or business stakeholder, with clearly labeled measures and filters rather than unnecessary visual complexity.

2.3 References

1. **Atul1804, Superstore Sales Dashboard (GitHub).** https://github.com/atul1804/Superstore-Sales-Dashboard
   Describes a Power BI Superstore project with sales overview, customer analysis, and profit analysis. Useful as a reference for report structure and interactive filtering. Its listed fields include Profit and Quantity, which are not present in our supplied CSV, so those measures cannot be reproduced from our current dataset.
2. **Mzaeemhaider, Superstore Sales Dashboard Power BI (GitHub).**
   https://github.com/mzaeemhaider/superstore-sales-dashboard-powerbi
   Describes an interactive Power BI project focused on sales trends, customer insights, regional performance, and executive KPIs. Relevant to our available date, customer, region, category, and sales fields.
3. **Microsoft Learn, Power BI documentation.** https://learn.microsoft.com/en-us/power-bi/
   Official product guidance for building reports, modeling data, and consuming insights.
4. **Microsoft Learn, Power Query documentation.** https://learn.microsoft.com/en-us/power-query/
   Official guidance for importing, transforming, and preparing data for analysis.

3. Requirements Gathering

3.1 Stakeholders

Stakeholder	Needs
Sales / store manager	Needs a concise view of overall sales and comparisons by time, region, and product grouping.
Sales or business analyst	Needs reliable fields, consistent calculations, and the ability to filter and compare results.
Project team	Needs clear responsibilities, agreed scope, and reproducible preparation and validation steps.
Trainer / evaluator	Needs evidence that the team followed the requested plan, reviewed relevant solutions, and defined testable requirements.

3.2 Functional Requirements

- FR-01: The solution shall import the supplied Superstore Sales Dataset.csv file into Power BI.
- FR-02: The solution shall inspect and prepare field types, especially Order Date, Ship Date, and Sales.
- FR-03: The solution shall provide a Total Sales measure based on the Sales field.
- FR-04: The solution shall summarize sales by category, sub-category, region, segment, ship mode, and time period.
- FR-05: The report shall allow users to filter results by available dimensions such as date, region, category, segment, and ship mode.
- FR-06: The report shall show the number of distinct orders and average sales per order using documented definitions.
- FR-07: The team shall validate displayed totals against the source dataset and document any data-quality decisions.
- FR-08: The team shall deliver the report and supporting documentation within the agreed project timeline.

3.3 Non-Functional Requirements

Requirement	Description
Usability	Visuals should have clear titles, readable labels, consistent formatting, and intuitive filters.
Performance	Common filters and visuals should respond smoothly on the team's available computer and the supplied dataset size.
Accuracy	Measures must use consistent aggregation; date fields must be parsed correctly; totals must be checked against the source.
Maintainability	Data transformation steps and measures should use understandable names and be documented so another team member can update them.
Privacy	Avoid exposing customer names or identifiers unless necessary for the analysis and permitted by the project context.
Compatibility	The final report should open in the Power BI environment available to the team; any version or licensing constraints should be checked before delivery.

3.4 Acceptance Criteria

- The CSV loads without losing the required columns.
- Order Date and Ship Date are recognized as dates, and Sales is numeric.
- The 11 missing Postal Code values are reviewed and handled transparently.
- Total Sales and distinct order count are validated against calculations from the source file.
- Category, region, time, segment, and shipping comparisons are available where relevant.
- Filters update the report consistently and all visuals have clear titles.
- The project documentation contains the three requested sections only.

3.5 Assumptions and Constraints

The supplied CSV is treated as the project's current source of truth. The file contains Sales but does not contain Profit, Quantity, Discount, or cost fields; therefore, profit margin, quantity sold, discount impact, and profitability KPIs are not included. The dataset's currency is not explicitly identified in the supplied file, so the report should not assign a currency symbol unless confirmed by the trainer or dataset source.